



Review

Evolution of Digital Twin Frameworks in Bridge Management: Review and Future Directions

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Abstract: Over the last decade, the digital twin (DT) concept has effectively revolutionized conventional bridge monitoring and management. Despite their overall success, current bridge DTs encounter conceptual ambiguities, hindering their inherent potential for practical implementation. Moreover, intelligent decision support models have not been properly considered as a component of the bridge DTs framework to enhance the reliability of decisions for asset maintenance. Therefore, this paper conducts a scientometric analysis and a comprehensive state-of-the-art review, exploring current bridge DT research trends and architectures and introducing an enhanced conceptual framework for bridge DTs. To this end, more than 480 research publications have been reviewed, compared, and analyzed. The research result encompasses the redevelopment of a multilayer DT framework, fostering its implementation in the full lifecycle of bridge infrastructure while exploring the potential integration of decision support systems and data fusion from advanced technologies to improve the overall efficiency of implementing DT technology in bridges.

Keywords: bridge engineering; digital twin; Decision Support System (DSS); digital infrastructure; digital transformation



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1. Introduction

Nowadays, the need for inspection and management of bridge infrastructures, as one of the most important components of the transportation network, has been significantly increased. Employing traditional inspection methods, which rely on on-site inspectors for regularly scheduled condition assessments, is quite laborious, expensive, and time-consuming. This approach leads to substantial budget allocations for the management and maintenance of the bridge infrastructure assets [1,2]. Moreover, these regular inspections have proven to be ineffective, as the bridge's condition constantly changes over time, and damages could appear after a periodic inspection and not be detected until the next one, which may result in further deterioration of the bridge and significant increased cost of its eventual repair or replacement, if not to its collapse, such as the recent catastrophes of the Caprioliola bridge in Italy in 2020 [3] and Mexico City railway bridge in 2021 [4].

To overcome the limitations of conventional inspection methods, the revolutionary Building Information Modeling (BIM) methodology is employed to provide a virtual model of the asset through the application of innovative technologies [5]. Over the last decade, application of Bridge Information Modeling (BrIM), as the specific form of BIM in the context of bridge engineering, has provided faster and more effective solutions in the bridge management processes using digital models [6]. The BrIM has emerged as a well-established platform for the sophisticated bridge management tasks by combination of bridge 3D models with other related data. It consists of geometric data in the form of a 3D Computer-Aided Design (CAD) model that is linked to non-geometric data, such as

on-site inspection reports, damage documentations, explanations of maintenance actions, the utilized construction/repair material, etc. [7]. This integration enables enhanced visual representations and superior methods for documenting bridge inspections, thus ensuring that this valuable information is readily available for any future use [8]. Over the years, several Bridge Management Systems (BMS) have been proposed by researchers which utilize BrIMs as their primary foundation for decision support through different stages of bridge lifecycle, including design [9], construction [10], maintenance, and management [11]. However, considering the time and effort involved in data collection and 3D model reconstruction, as the essential parts of BrIMs, there is a need for an efficient management strategy to record the extensive condition information of the bridges, which are often fragmented or incomplete [12]. Moreover, BrIM has limitations in dealing with ever-expanding data collection methods and does not consider means for mutual interactions with the physical asset, which results in the inability to provide real-time feedback on the state of asset operations or make enhanced decisions regarding maintenance strategies [13].

Digital twins (DTs) have been proposed to enhance the functionality of BrIM through digital transformation of the entire bridge lifecycle and have become increasingly popular for bridge infrastructure management [14]. DT comprises a computational model and a real system that is employed for monitoring in-operation bridges based on multiple sources of data, controlling the operation functionalities, and optimizing the efficiency of the asset [15]. Moreover, DTs support data exchange between physical and digital objects in both directions and consist of a virtual representation of a real-world asset and a connectivity module, which synchronizes both the physical and virtual assets along with the bridge's lifecycle stages. Several approaches, such as big data analysis, machine learning, and cloud/edge computing, have been used in DT platforms to manage the increasing input data [16].

DTs can also be employed for bridge maintenance purposes through accurate data analysis and offering solutions prior to any breakdown [17]. This capability can be achieved by constant and ideally real-time monitoring, which proves to be beneficial in identifying any defects or problems at an early stage [18]. As DT represents a virtual replica of the physical asset through the BrIM approach, crucial functional areas of the bridge could be assessed during a bridge health monitoring process to generate information on a variety of performance characteristics, such as environmental conditions, loads, and reactions of the structure to those loads. These data could be subsequently applied to the processing system for further structural analysis both at local-element and global bridge scales [14]. By analyzing these data, DTs may recognize potential issues in a system. Consequently, the bridge management team or other interested parties can employ this valuable information to make predictive maintenance decisions, thereby enhancing reliability, increasing efficiency, and optimizing the accessible resources at their disposal [19]. In addition to online monitoring and predictive maintenance, a complete lifecycle management of the infrastructure from design, development, manufacturing, and maintenance is provided by DTs, which is beneficial in optimizing the lifecycle and improving the overall quality of the operation.

Since 2019, there has been rapid progress in bridge DTs, and several review papers have delved into the advancement, key technologies, and reference models of bridge DT. Honghong et al. [12] conducted a review comparing and analyzing 116 documents on BIM and DT, suggesting a DT-enhanced BIM framework to promote comprehensive digital bridge engineering throughout its lifecycle. Rios et al. [20] reviewed a total of 76 relevant studies from 2017 to 2022, identifying gaps such as software interoperability, enhancing anomaly-detection algorithms, and defining fusion approaches for DTs at the macro-scale. Hosamo et al. [21] carried out a review focusing on papers from 2017 to 2022, underscoring information standardization as a notable obstacle that needs to be addressed for the construction sector to fully leverage DT utilization. Moreover, various articles have explored the wider application of bridge DTs. Pregolato et al. [22] provided a state-of-the-art review of DTs in Civil Engineering, implementing their suggested framework in a bridge

case study. Mihai et al. [23] reviewed key enabling technologies, challenges, and prospects of DTs, offering detailed insights into applications and case studies in bridge management.

Although the concept of DT and the critical value of its integration into bridge engineering has been investigated in many research studies, the current implementation of DTs in bridge management is still at a nascent stage [12,22]. This causes different classifications of bridge DTs from different perspectives, such as level of details [24], maturity [12], level of complexity [25], and data connectivity [26]. Moreover, researchers have introduced various mixed definitions of bridge DT, which makes it difficult to establish an evolved digital platform with seamless integration of technologies [12,27]. Many methods in the literature define bridge DT as a FEM updating approach that collects data from different sensors, explores potential relationships between different datasets, considers uncertainties, and makes predictions for bridge safety and reliability in specific situations [28–32]. Other methods consider DTs as an as-is BrIM updating approach to generate a DT-enhanced BrIM framework with the latest inspection damage information [33–36]. These research studies can help in tackling the dilemma of changing from static and enclosed information with recurring compatibility difficulties to a connected data model, where the physical asset can be completely represented as a DT. Other methods are based on the BrIM platform that consider the acquired 3D model, FEM and information integration, and data exchange [27,37–39].

These research studies indicate the presence of conceptual ambiguity in the current research related to bridge DTs. This ambiguity manifests in varying concepts, tools, and application examples, making it challenging to define a standardized digital workflow for implementing DT in bridge management tasks. Concerning the design of the current bridge DTs and from an operational point of view, there are two key concerns that have not been adequately explored in previous studies:

- **Developing an evolved conceptual framework of bridge digital twin:** As the concept of bridge DT includes a wide range of sophisticated capabilities, varying from a simple digital representation of bridge to complicated models with predictive capabilities [14], various mixed definitions/applications have been developed in the literature. Although some attempts have been conducted to establish a lifecycle DT framework [12], there is still a need for an evolved multilayer conceptual framework of DT for proactive bridge monitoring. Therefore, the development of a conceptual framework for DTs is required by defining the basic components and their interconnections.
- **Developing intelligent decision support systems in digital twins:** At the fundamental part of the maintenance-oriented bridge DT, there exists a decision support model, which reflects the pertinent operations and knowledge in the process of bridge management and acts as a vital component of the virtual twin [8,40]. The accuracy and efficiency of inferences extracted based on the decision support model directly impact the reliability of decisions for asset maintenance [15]. Despite the extensive research conducted in this field, it remains challenging to make practical yet intelligent decisions for decision-makers without the power of DTs, as it requires extensive data analysis and alignment of several objectives across different agencies [41].

Therefore, this research study provides a comprehensive review of DT from the bridge management and maintenance perspective to provide an overview of the state-of-the-art DT components and classify them according to their objectives, functionalities, and integrated technologies for bridge engineering. Thereafter, based on the investigation results and research gaps, this paper proposes an evolved multilayer conceptual framework of DTs with an optimized management plan supported by decision support systems, which could overcome the inherent vulnerability of human subjectivity in the planning process due to varying levels of experience and expertise of the involved decision-makers.

In this research, as explained in Section 2, a total of 2300 research papers spanning over the last 5 years were collected and analyzed through a systematic literature review methodology to highlight the research gaps in the existing literature. Subsequently, the developments in DTs, including virtual models, data collection, and data connection

methods (Section 3), along with classifications and integrated technologies (Section 4), were summarized through an in-depth discussion. Thereafter, an evolved multilayer DT framework has been proposed to promote the full lifecycle implementation of DTs for bridge infrastructures.

2. Literature Analysis Methodology

The current research study employed a quantitative analysis to evaluate the published literature concerning DTs for bridge health monitoring and management. Initially, bibliometric techniques were employed to analyze the published literature, to provide a roadmap from the research structure, and to develop different concerns/objectives using extensive academic datasets. Thereafter, a comprehensive analysis of the intellectual landscape of the current research areas in bridge management DT, along with network modeling and visualization, was presented, which will enable readers to obtain a comprehensive understanding of research trends and highlight the knowledge gaps in this field of study.

Moreover, keywords and abstracts were analyzed to pinpoint influential researchers shaping this field of study and to obtain research patterns through the analysis of keyword co-occurrence and co-authorship using well-known VOSviewer (version 1.6.19) and CiteSpace (version 6.2.6) software.

2.1. Literature Search Strategy

For the literature search, a complementary search of relevant scholarly articles was undertaken in Google Scholar, Scopus, and Web of Science. An overview of the literature search strategy is shown in Figure 1. At first, queries including keywords (Digital Twin, bridge/structural health monitoring (SHM), Internet of Things (IoT), Artificial Intelligence (AI), laser scanning, UAV photogrammetry, Ground-Penetrating Radar, point-cloud processing, infrared thermography, hyper-spectral image, and decision support) were used, considering words with similar meaning and common variations and combinations.

The search returned a total of 2303 research papers originating from different academic fields, including physics and astronomy, materials science, computer science, and mathematics, which were not pertinent to the research topic. Consequently, two stages of screening were implemented to refine the initial outcomes. Initially, exclusion criteria based on subject area were employed to filter out the irrelevant articles, followed by a restriction on the publication year, limiting it to the most recent five years (2019 to 2023) to capture recent developments and maintain a manageable scope for the review. The first screening stage resulted in the exclusion of 534 papers, whereas the subsequent stage eliminated an additional 356 articles.

After this stage, the process of removing duplicate records, filtering, and eligibility evaluation for all the remaining papers was carried out. Therefore, based on the DOI numbers, all duplicated articles and those lacking a valid DOI number were excluded, which resulted in 237 papers being discarded. Again, the remaining papers were analyzed during another eligibility determination stage to include specific research papers relevant to the scope of this paper. To this end, the abstract and keywords of each article were carefully reviewed to ensure their compatibility with the aim of this paper. At this juncture, a further 695 papers were eliminated. Consequently, a total of 481 articles remained for the reviewing process. Considering the multiple objectives of the papers in this field, as illustrated in Figure 1, documents were divided into five subdivisions related to the fundamental components of DTs in bridge engineering, including data collection methods, data connection methods and integrated technologies, DT development research, and decision support/management in DTs. This categorization serves to highlight the prevalence of research papers in each area concerning the fundamental components of DTs in bridge engineering.

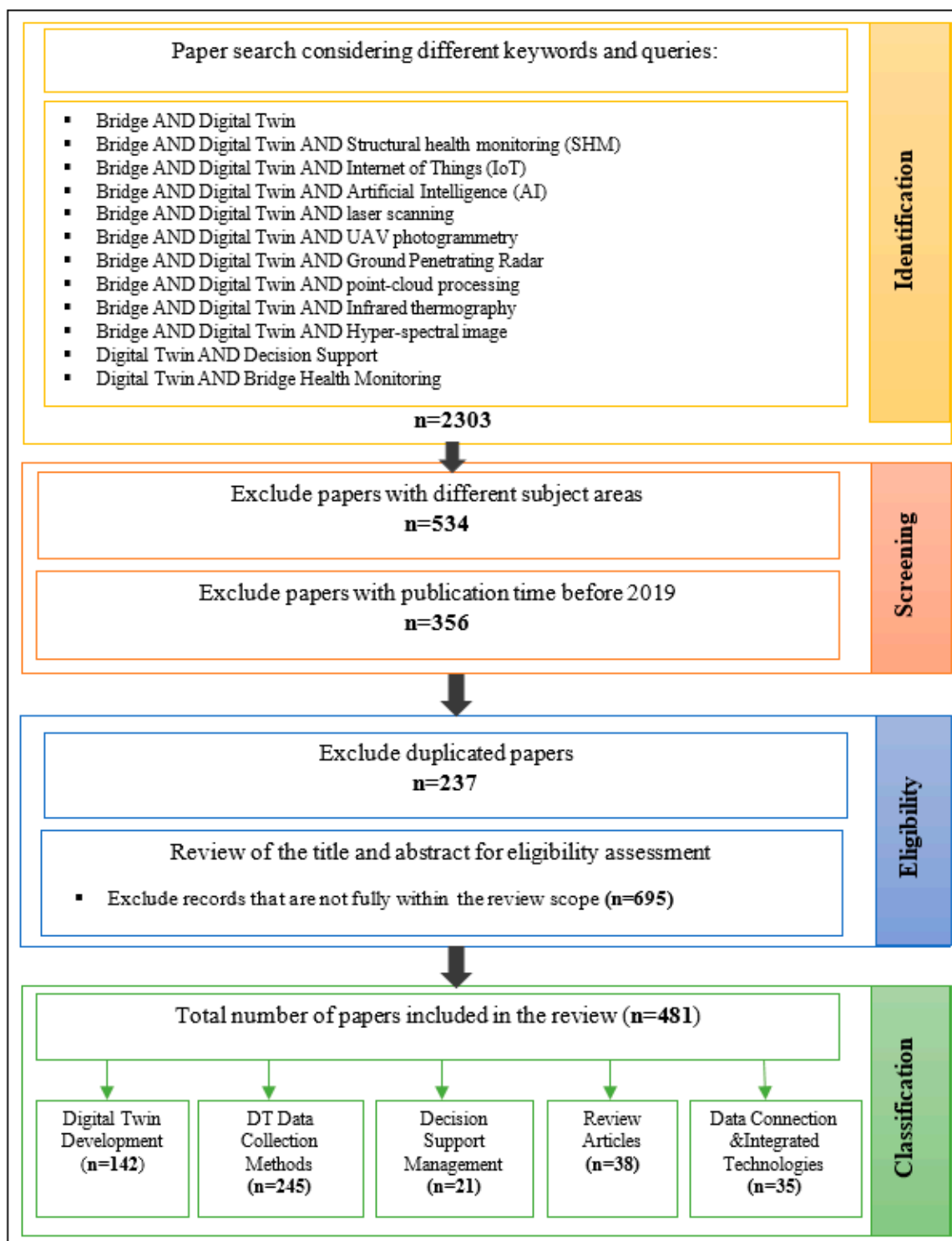


Figure 1. Flowchart of the literature analysis process.

The topmost reputable journals and conferences were identified in the reviewing process and the results are highlighted in Table 1. As it shows, *Automation in Construction* and *Remote Sensing* were the primary sources for published papers concerning the bridge DT.

Table 1. Top journals and conferences according to the published papers from 2019 to 2023.

Journal Title	Number of Published Papers	% of Total Included Publications
Automation in Construction	39	9%
Sensors	29	7%
Remote Sensing	22	5%
Structure and Infrastructure Engineering	18	4%
Applied Sciences	15	3.5%
Engineering Structures	10	2%
Construction and Building Materials	9	2%
Journal of Bridge Engineering	8	2%
Buildings	8	2%

Conference Title	Number of Published Papers	% of Total Included Publications
Lecture Notes in Civil Engineering	22	5%
IABSE Symposium/Congress	10	2%
Computing in Civil Engineering	9	2%
Conference on Bridge Maintenance, Safety, and Management, IABMAS	4	1%

2.2. Co-Occurrence of Keywords Analysis

The analysis of co-occurrence demonstrated the frequency of keywords in the literature and articulated the underlying concepts, concerns, and subjects explored in the published literature. Furthermore, it provided a concise overview of the current research frontiers. VOSviewer was used to extract the co-occurrence of keywords in bridge DT, and the outcomes are visually depicted in Figure 2, displaying a wide range of keywords that emerged in the literature as a result of extensive studies in this area. Table 2 also presents the details of the keywords, such as number of occurrences and total link strength.

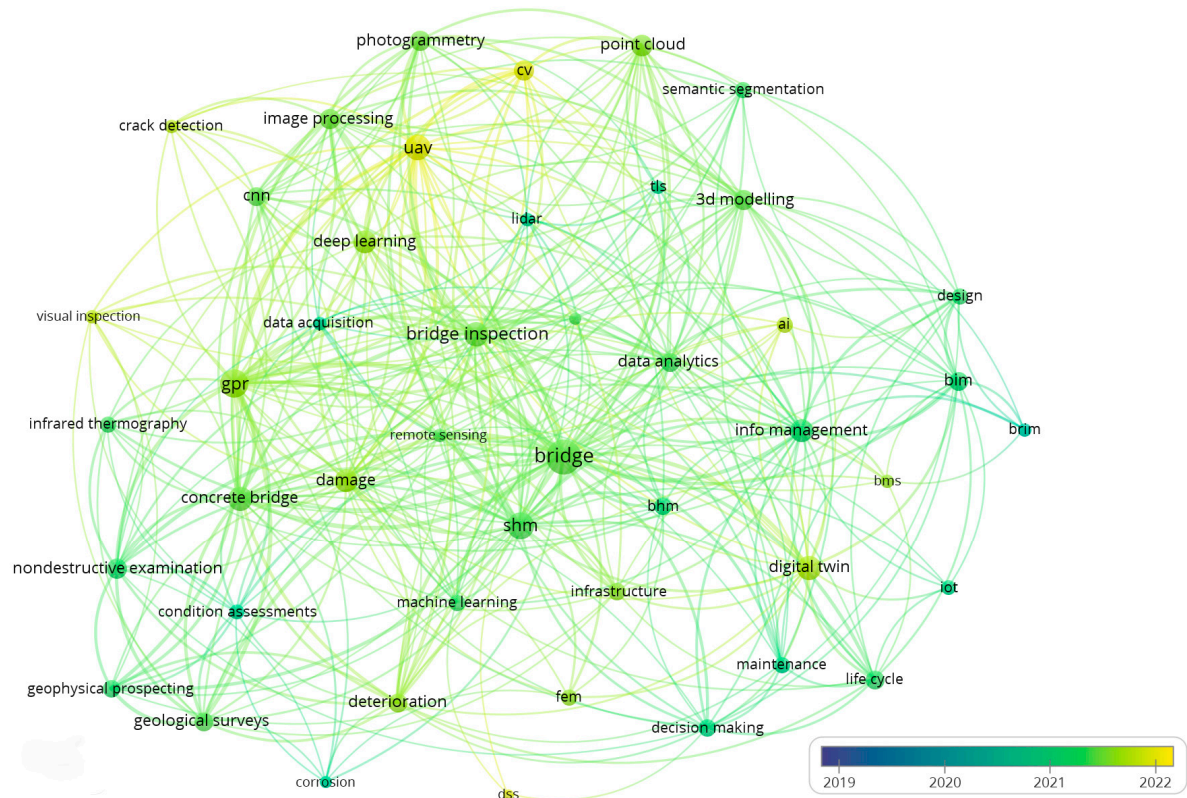


Figure 2. Network of co-occurring keywords.

Table 2. List of keywords and related network data.

Classification	Keyword	Occurrence	Total Link Strength	% of Total Keyword Occurrence
Digital Twin Development	Bridge	220	1103	10%
	Structural Health Monitoring (SHM)	110	513	5%
	Bridge Inspection	85	503	4%
	Concrete Bridge	80	459	4%
	Digital Twin	80	290	4%
	Information Management	76	408	3%
	3D Modeling	53	314	2%
	Data Analytics	48	280	2%
	Lifecycle	43	223	2%
	Bridge Health Monitoring (BHM)	39	175	2%
	Infrastructure	39	213	2%
	Design	33	181	1%
	Finite Element Method (FEM)	31	122	1%
	Maintenance	30	162	1%
	Condition Assessments	25	184	1%
	Bridge Management System	18	117	1%
	Visual Inspection	18	116	1%
	Automation	15	99	1%
DT Data Collection Methods	Ground Penetration Radar (GPR)	126	764	6%
	Unmanned Aerial Vehicles (UAV)	93	506	4%
	Damage Detection	71	399	3%
	Point Cloud	60	259	3%
	Computer Vision (CV)	57	271	3%
	Image Processing	54	293	2%
	Photogrammetry	53	284	2%
	Nondestructive Examination	50	314	2%
	Deterioration Detection	48	322	2%
	Geological Surveys	47	327	2%
	Geophysical Prospecting	35	261	2%
	Infrared Thermography	31	160	1%
	Semantic Segmentation	30	135	1%
	Crack Detection	22	113	1%
	Data Acquisition	21	124	1%
	Lidar	20	126	1%
	Remote Sensing	18	95	1%
	Terrestrial Laser Scanning	18	120	1%
	Corrosion Detection	16	91	1%
Data Connection and Integrated Technologies	Deep Learning	70	321	3%
	BIM	46	218	2%
	Convolutional Neural Networks (CNN)	45	221	2%
	Machine Learning	29	143	1%
	Artificial Intelligence (AI)	28	124	1%
	IoT	24	79	1%
	BrIM	21	99	1%
Decision Support Management	Decision Making	38	210	2%
	Decision Support System (DSS)	13	73	1%

Based on this information, it can be determined that the predominant keywords found within the literature concerning digital twin development were “structural health monitoring” and “bridge inspection”. Regarding the data collection methods for DTs, the keywords “GPR” and “UAV” frequently occurred. In terms of data connection and integrated technologies, the keywords “Deep learning” and “BIM” frequently emerged.

Conversely, the frequency of occurrence for other keywords, such as “IoT”, “BrIM”, and “Decision Support System (DSS)”, was notably lower than other keywords, which

indicates the research endeavors associated with these topics in bridge management are limited. Hence, there is a significant need for further investigations in these fields.

2.3. Network of Countries and Institutions

In order to illustrate the contributions of various countries in the development of bridge DT, a network was constructed using the well-known CiteSpace (version 6.2.6) software, which depicted the distribution of research publications. Figure 3 displays the primary contributors to the publications in this research field, such as China (with 121 papers), the United States (with 118 papers), the United Kingdom (with 38 papers), Italy (with 36 papers), South Korea (with 30 papers), and Australia (with 29 papers).

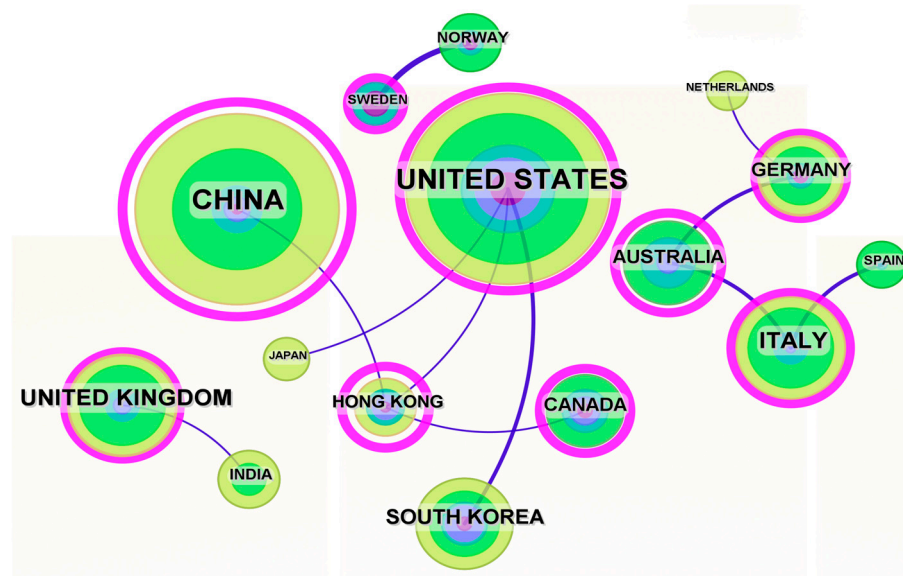


Figure 3. Network of countries/regions.

3. Review of Bridge Digital Twin Components

According to the literature and the reference framework proposed by Tao et al. [42], the ideal bridge DT encompasses five main primary components, as shown in Figure 4, including physical entity (bridge), virtual model, data collection, data connection/integration, and user as the final operator and manager of the bridge [12,22,43]. In the context of bridge DT, the physical entity of the bridge is regarded as the service objective. This entity can be considered either as the entire bridge structure with various types and styles with massive information or as separate components classified based on their length and position [12]. Therefore, in this section, the current research methods in the four remaining components will be reviewed.

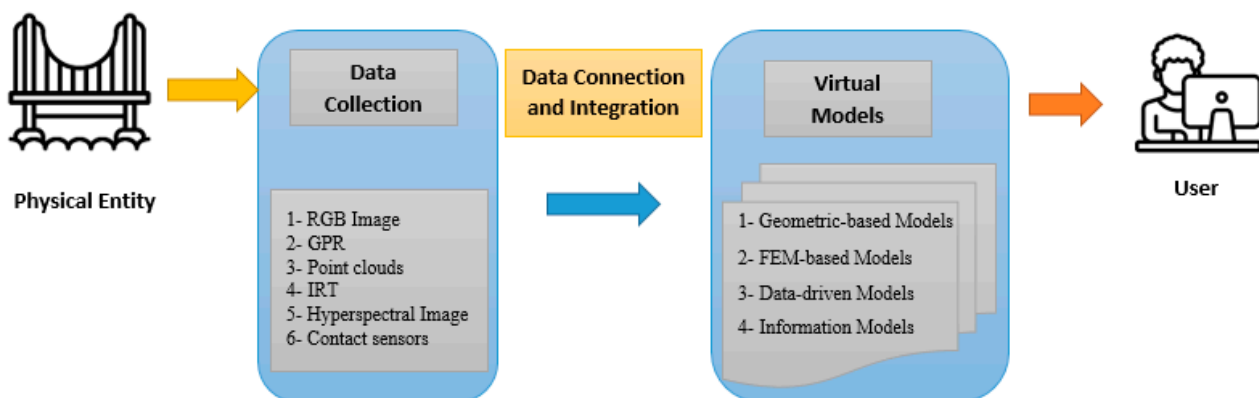


Figure 4. The primary components of ideal bridge digital twin.

3.1. Digital Twin Virtual Models

The generation of the virtual model for the bridge, serving as the main skeleton of the digital twin (DT), emerged as a critical and intricate phase aiming to deliver precise outputs and feedback of adequate quality and quantity. The quality of virtual models and their close resemblance to the physical model plays an important role in the implementation of DT for applications in bridge engineering [44]. In this regard, several methods have been proposed by researchers to build virtual models in bridge DTs, which can be categorized into four groups based on their components and functionalities. The main functions have been summarized in the following subsections.

3.1.1. Geometric-Based Models

The 3D surface reconstruction models in bridge DT provide an accurate as-is model from the surface state of the bridge, which helps in determining changes in bridge geometry, defect detection, and overall condition assessment [12]. Based on the literature, two methods have been widely used by researchers to generate 3D models from the bridge surface in DTs, including laser scanning and Unmanned Aerial Vehicles (UAV) photogrammetry.

- **Laser Scanning**

The employment of laser scanning has become widely recognized as a non-contact measuring technique for rapid and precise data collection from an object's surface [45]. Laser scanning systems can be classified into two categories, namely aerial and terrestrial, based on the positioning of the laser sensor during the process of data acquisition [14].

Unlike traditional surveying methods, Terrestrial Laser Scanner (TLS) technologies can capture the details of the entire scene and its color attributes, which exhibit their potential for a variety of DT applications, owing to their sub-millimeter precision, rapid speed, and cost-effectiveness [46]. Several studies used TLS to generate DTs of the bridges. For instance, Hu et al. [34] proposed a method for 3D model generation using point cloud data acquired from TLS. Omer et al. [47] also applied TLS for the purpose of constructing the DT model to examine the condition of concrete bridge structures and employed virtual reality (VR) for the inspection of bridges within an immersive 3D virtual environment. Similar studies were also carried out using TLS concerning the development of an as-built model for structural assessment of bridges, which could be used as a reliable basis for 3D model generation of bridge DTs [48–50].

Aerial laser scanning in the form of Light Detection and Ranging (LiDAR) is another method for 3D modeling of bridges. Aircraft-mounted laser scanning can increase scanning rates and accuracy, leading to cost and time savings when scanning large bridges. As an example of practical application, Lugo et al. [51] proposed an approach to create an accurate virtual model from a pedestrian bridge using LiDAR scanning. In another study, Abdel-Maksoud et al. [52] employed a UAV-based LiDAR technology to quickly capture georeferenced 3D models for bridge inspections.

- **UAV Photogrammetry**

With the rapid development of non-metric digital cameras mounted on drones, the UAV photogrammetry has been leveraged as a practical and efficient technology to construct the bridge virtual twins [53].

In UAV-based photogrammetry, a pre-designed flight plan is employed to capture high-resolution aerial images remotely from various perspectives of the bridge. Subsequently, a 3D point cloud is generated through additional post-processing techniques, utilizing matched key points via well-known Structure from Motion (SfM) or Multi-View Stereopsis (MVS) algorithms [54]. Therefore, the acquired 3D model represents the virtual twin of the bridge, and quality assessments based on structural information can be applied to the reconstructed model.

Several scholarly research studies in the literature have underscored the significant advantage of employing UAV photogrammetry as a reliable technique for bridge inspection, surface evaluation, and DT generation [55]. For example, Chen et al. [56] proposed a

procedure for bridge inspection that utilizes UAV-based photogrammetry to reconstruct 3D models and subsequently enable virtual inspection and damage detection. Following a similar approach, Pan et al. [57] introduced a semi-automated algorithm to extract a three-dimensional digital representation of a bridge using UAV-based photogrammetry, and they assessed the efficacy of this approach through a case study of an existing heritage bridge in China. Jalinoos et al. [58] used the UAVs to generate 3D DT to assess the post-hazard damages of highway bridges. Mirzazadeh et al. [59] developed a framework for bridge inspection using UAV-derived 3D models.

Although UAV photogrammetry has been considered as a popular option for the digital modeling of bridges, several concerns in the UAV photogrammetry pipeline, including the path planning to cover blind spots [60], the quality of image matching process [61], the accuracy of image orientation [62], and the geometric accuracy of reconstructed models [54], require to be effectively addressed. In addition, as the two techniques of TLS and UAV photogrammetry have been widely used for DT generation, several studies compared the accuracy of these two methods [63,64].

In addition to these reconstruction methods that generate as-is models from surface data using point clouds, forward geometric modeling methods facilitated by commercial 3D modeling software also serve as valuable tools for creating detailed 3D models of bridge components. Their primary focus lies in predicting future geometric changes or designing specific aspects of a structure, rather than providing an accurate representation of the bridge's current state based on surface data.

3.1.2. Finite Element Method (FEM) Models

The FEM-based virtual model has been considered as the most prominent numerical technique for bridge DT generation to analyze the structural condition [65]. The virtual model is simply generated using a validated FEM model with commercial FEM software that can not only output mechanical properties, such as displacement or mechanical load, but also facilitate the direct generation of the electric output originating from the sensors that are intricately embedded within the structure, such as electric charges or voltages [14].

As an example, Cervenka et al. [66] calibrated a FEM-based DT of the bridge by applying the degradation models derived from chemo-mechanical computational methods. Lai et al. [67] proposed a DT-based method for bridge structural health monitoring using the combination of the FE method and non-destructive testing (NDT) sensor technology. Moreover, Lin et al. [31] proposed a DT-based method for collapse fragility assessment of long-span cable-stayed bridges under strong earthquakes based on three finite element models of the bridge.

The current FEM-based models pose challenges in achieving an accurate visualization of bridges because they often rely on oversimplified approaches, such as linear assumptions. These simplifications can introduce biases into the output and result in unrealistic simulations. Additionally, these models might be unsuitable for real-time simulation scenarios due to their demand for substantial processing resources [14].

3.1.3. Data-Driven Models

Data-driven simulation models are generated based on the validated data obtained from experiments/measurements on the actual structure and produce a model with lower complexity to analyze the status of the bridge quickly. The main aim of data-driven models is to provide a reliable alternative to intricate models by minimizing the disparity between the high-dimensional model and its simplified counterpart. Artificial Intelligence (AI)-enhanced algorithms are extensively adopted in bridge DT approaches to achieve this objective [25,68].

In data-driven models, data from a physical bridge are collected using emerging sensor technologies and IoT (Internet of Things) devices. These data are then applied to big data analytics and advanced AI methods used to simulate the exact functionality of the bridge [16]. Therefore, the bridge DT would be able to make dynamic decisions based on

the physical sensors. Consequently, real-time data collection becomes an important part of data-driven DT, which is facilitated by the deployment of IoT devices.

Anomaly detection techniques can be employed within data-driven models to diagnose bridge damages by detecting abnormal data or behavior [69]. As an example, a learning-based data anomaly detection method was proposed by Bao et al. [70] for bridge condition assessment. Moreover, a review of DT-based anomaly detection systems is presented in [20].

Sensor-based data-driven DT is another form proposed by Meixedo et al. [71] for a railway bridge to simulate baseline and damage scenarios realistically. As a practical instance, a data-driven DT framework was developed at the London DT Research Centre that comprises four components, including sensors, a cloud layer for data storage, a machine learning-based data analytic, and a web application to visualize the results [23].

3.1.4. Information Models

The information models in the context of bridge DT are systems that store and organize data related to various aspects of the bridge during its lifecycle. This includes information about the entire lifecycle of the bridge, from its initial design and construction through its ongoing service life and maintenance [40]. The information model can also describe the knowledge that has been derived from the analysis of historical data, expert decisions, and predetermined logic. This information equips the virtual model with the ability to reason, judge, and extract rules from the bridge lifecycle and management plans [72]. The rule extraction process involves the utilization of both symbolic methods, such as decision tree and rough set theory, as well as connectionist methods, such as neural networks. The description of rules employs various methods, such as semantic web notation and XML-based and ontology-based representation [73]. These rules can then be updated based on feedback obtained from the application process and periodic evolutions.

As an example, Giorgadze et al. [74] proposed a conceptual model for the DT integrated with lifecycle information of the bridge. Moreover, Kang et al. [75] proposed the use of multimedia knowledge-based bridge health monitoring using DTs to synchronize real and virtual spaces, analyze bridge situations, predict the maintenance time, and reduce maintenance costs while extending the bridge's lifespan.

3.2. Digital Twin Data Collection Methods

Effective data collection methods play a vital role in DT development, offering valuable information about different aspects of the physical object's performance. Different methods have been used to collect data for DT, which are discussed in the following sections.

3.2.1. RGB Image

In recent years, with the rapid development of computer vision and machine learning technologies, damage detection based on images is widely used across all types of infrastructures, and many AI-based approaches have been proposed to collect data for bridge DTs. Therefore, an overview of image-oriented damage detection methods is provided in this section.

- **Crack Detection**

Crack detection on the surface of concrete structures is regarded as a crucial element in evaluating structural integrity. The presence of cracks on the surface of concrete is a primary imperfection in civil infrastructure, resulting in detrimental effects on the durability, safety, and overall integrity of concrete structures [76].

The first part of crack detection methods aims to recognize cracks and extract the crack boundary from images. Therefore, numerous approaches have been developed as effective tools to tackle the challenges encountered in crack detection in practical applications. In the early stages of research on vision-based automated crack detection, the main emphasis was on developing algorithms for detecting crack edges using traditional methods of digital image processing. However, these approaches have a major drawback of highlighting local

patterns more than global features, even though cracks are part of the whole image. As a result, some cracks in an image may be overlooked if more emphasis shifts toward local patterns in the image [77].

To address this challenge, several investigations have started incorporating machine learning techniques with image processing methods in order to enhance the accuracy of crack detection. The combination of machine learning and feature extraction based on image processing has the potential to improve the accuracy of crack detection. Nonetheless, it might still struggle to detect cracks in images exhibiting complex background characteristics and noise. The main underlying issue stems from the manual extraction of crack features. While manually crafted crack features could be highly effective for a specific dataset, their performance might degrade when applied to a new set of crack images captured from a more complex environment.

Recently, deep learning, as a subfield of machine learning, has seen rapid development due to its ability to combine automatic feature extraction and nonlinear classification. Numerous predictive models featuring deeper network architectures have been proposed for the detection of surface cracks. For example, Li et al. [78] made modifications to AlexNet in order to create a deep CNN model for concrete crack detection. Their CNN method was incorporated into mobile devices for bridge inspection purposes and achieved successful results on concrete surfaces without being affected by noise. Jo et al. [79] developed an independent crack classification system that relied on a deep belief network. This network was trained using a dataset consisting of 15,000 infrared and RGB images, some of which contained cracks while others did not.

Yu et al. [77] proposed a vision-based automated crack detection method in concrete structures using pre-trained CNNs, transfer learning, and decision-level image fusion. In their approach, each individual pre-trained CNN model undergoes fine-tuning for an initial assessment of surface conditions using transfer learning. Following this, a modified Dempster–Shafer (D-S) fusion algorithm was devised to integrate the initial evaluation outcomes of structural surfaces. Ultimately, with decision-level image fusion, there was a significant improvement in the precision and confidence level of the prediction results. Furthermore, Ding et al. [80] developed a UAV-based to precisely detect the concrete cracks using the full-field-scale calibration of the gimbal camera to localize the cracks and a learning-based boundary refinement transformer method for crack segmentation in acquired images. A detailed comparison of the learning-based algorithms for defect classification, segmentation, and detection, and related datasets can be found in [81].

Furthermore, the technique of 3D mapping can be integrated with the crack detection map at the pixel level to precisely locate the cracks in the 3D space. Different studies were conducted in this regard. As an example, Zhao et al. [82] proposed a system for the identification and localization of damages in concrete dams. The presented system integrates the YOLOv5s-HSC algorithm with a 3D photogrammetric reconstruction method, which allows for the accurate detection and precise localization of damages in concrete dams. Kim et al. [83] developed a framework for the use of stereovision with wide-angle images in concrete bridges. This framework facilitates precise crack quantification, as well as efficient 3D reconstruction and depth estimation. Recently, Deng et al. [84] developed a framework for the 3D reconstruction and measurement of cracks in concrete structures using binocular videos. The suggested framework introduces a method for 3D crack reconstructing, which is combined with a high-performance segmentation network and visual simultaneous localization and mapping (VSLAM). This approach relies on processing steps to accurately identify individual cracks within the overall structure and measure their lengths.

- **Corrosion Detection**

Image-based detection of corrosion and associated damages helps to overcome the challenges of manual inspection methods and reduces the expenses linked with damages caused by corrosion. Nevertheless, there has been limited exploration into vision-based

automated techniques for detecting structural corrosion and coating defects, despite their potential benefits, such as real-time processing and cost-effectiveness [85].

Coarse texture and the color are considered as the two primary visual attributes of corrosion in vision-based analysis. Consequently, most algorithms use these two main characteristics for the identification of corrosion, which may be employed either independently or integrated into a pattern recognition technique [86]. For example, Hoang et al. [87] have proposed a vision-based approach for the identification of pitting corrosion through the utilization of the history-based adaptive differential evolution with linear population size reduction (LSHADE) metaheuristic, the support vector machine (SVM), machine-learning, and image-processing methods. The performance of such algorithms is highly sensitive to the selected features in the defect area, which can be subjective and dependent on the operators.

Recently, several learning-based technologies that incorporate feature extraction and pattern classification components within a learning framework have emerged for the detection of structural corrosion and defects, leveraging ANNs and CNNs [88,89]. Zhao et al. [90] employed a deep learning approach to construct predictive models for assessing coating defects, comparing the performance of VGG19 and ResNet50 for this task.

Although deep-learning-based methods have shown superiority over traditional machine learning models in defect detection in various studies, accurately quantifying and distinguishing the severity levels of corrosion and coating defects remains challenging. This difficulty arises due to the complex relationship between defect generation mechanisms and their observable manifestations [91].

3.2.2. Ground-Penetrating Radar

According to the literature, Ground-Penetrating Radar (GPR) is one of the most widely studied non-contact testing technologies [92]. The fast data acquisition specification of GPR enables the collection of data without traffic congestion, which is significantly important for highways and bridges that experience heavy traffic flow. Consequently, as a non-destructive and quick electromagnetic inspection method, it has the potential to be employed in the determination of internal flaws within bridge decks and the arrangement of reinforcements by measuring the attenuation of reflected waves [93].

Different defects can occur within the bridge elements, and GPR could be helpful to detect these defects for bridge monitoring purposes. GPR can locate the defects in the bridge by detecting concrete delamination and rebar corrosion. In GPR data processing, rebar detection and localization through signal processing to extract rebar and non-rebar regions is an important task [94]. Several research studies have been conducted to extract the rebar's location. For instance, Dinh et al. [95] proposed an automated GPR analysis as an amplitude map to show the rebar's location and areas with corrosive environments. In another study, Asadi et al. [96] proposed a computer-vision-based method to detect rebars in GPR images obtained from highly deteriorated concrete bridge decks. Ahmadvand et al. [97] evaluated concrete bridge decks by employing CNN and GPR data, which were collected from experimental specimens, consisting of both defect-free samples and subsurface defects that were manually created. To evaluate the concrete bridges, another method employing GPR was proposed by Kumar et al. [98] to identify the properties of in situ material, including the raw average amplitude, the amplitude of reflected waves, and intensity across different periods or areas. These properties were then applied to a random forest model in order to forecast the compressive strength, density, and porosity of the material.

To mitigate the influence of expert knowledge on the determination of the actual subsurface conditions derived from GPR devices, machine learning algorithms have been developed to discern the patterns caused by internal defects [99]. For instance, Hu et al. [100] introduced an automated workflow for generating a map of bridge deck deterioration through the utilization of GPR scans. This workflow comprises three distinct steps. Initially, a Random Forest classification model is trained to identify regions containing rebar within

the GPR scans by employing Histogram of Oriented Gradients (HOG) features. Subsequently, a robust hyperbolic fitting technique is employed to fit each hyperbolic signature within the rebar region accurately. Lastly, the detected rebars, along with their respective locations and depth-corrected amplitudes, are utilized to generate the deterioration map. Furthermore, Zhang et al. [94] proposed a technique for the identification of corrosive conditions in bridge decks using GPR data, employing the single-shot multi-box detector (SSD) model. This approach can be divided into three distinct stages: data pre-processing, automatic rebar selection, and corrosive environment mapping. Initially, the GPR data are subjected to pre-processing to enhance the contrast of the hyperbolic characteristic in GPR scans. Subsequently, the rebar elements in the scan images are automatically identified utilizing the trained SSD model. Finally, the depiction of the corrosive environment on the bridge deck is achieved through the creation of a contour map of the bridge deck corrosion environment extracted from the reflection amplitudes of the rebars. In Li et al.'s [101] study, the YOLO v3 algorithm was utilized to identify the reflected signals from the reinforcement in the scanned images, and subsequently extract an approximate range of the thickness of the overburden. Subsequently, these derived cover thicknesses were imported into a CNN, accompanied by the EMI (electromechanical impedance) data, to simultaneously estimate the cover thickness as well as the diameter of the reinforcement.

3.2.3. Point Clouds

Point cloud data, which are enriched with a variety of information, including 3D coordinates, colors, and intensity values, is one of the most widely used tools to achieve full information about the damages of the bridge during the bridge's operation and maintenance cycle [1]. Point clouds are normally obtained through laser scanning or the photogrammetric 3D reconstruction process. Based on the 3D information obtained, geometric quality inspection and bridge maintenance management can be obtained at various stages [102]. Therefore, several point cloud processing algorithms have been introduced in order to effectively analyze the point cloud data.

As point cloud data are extracted from the entire structure of the bridge, discerning local damage from these data proves to be a challenging task. Therefore, some research studies concentrate on autonomously extracting the distinct structural elements of bridges [103]. Learning-based point cloud segmentation methods were among the most widely applied methods in order to segment the target components of the bridge from the point cloud data for further damage analysis. To this end, different point cloud processing networks, such as PointNet [104], have been proposed in the literature to segment the large-scale point cloud data.

As an example, a deep learning method was proposed by Kim et al. [105] to extract the bridge components using the PointNet processing network and consensus labels. In another study, a graph-based CNN was proposed by Lee et al. [106] to achieve a more precise and authentic model of railway bridges that are equipped with electric poles. In their model, detailed local characteristics are acquired through the incremental consideration of neighboring points while maintaining a constant total number of neighbors. Furthermore, Xia et al. [107] proposed a methodology for the automatic semantic segmentation of bridge point clouds. Their methodology relies on machine learning techniques and multi-scale local descriptors, which are calculated for each point in the point cloud. These descriptors are then used to train a deep classification neural network. In the study conducted by Perry et al. [108], the damage size and location information are recognized by the combination of image data, point cloud, and machine learning techniques to detect damage and monitor the progression of bridge cracks.

Creating or updating as-is BrIM models for existing bridges is another application of point cloud data. As an example, Zhao et al. [109] generated a BrIM from laser scan point cloud obtained from a concrete bridge. A framework was also developed by Turkan et al. [110] to speed up the generation of BrIMs from UAV photogrammetry point

clouds. Mohammadi et al. [7] proposed a sliced-based approach for TLS-derived BrIM generation for a cable-styled bridge.

As point clouds could be generated using UAV photogrammetry and laser scanning technologies, some studies, such as [56,64], compared the accuracy of the final point cloud and provided advantages and disadvantages of both techniques for structural health monitoring of bridges.

3.2.4. Infrared Thermography

Infrared thermography (IRT) is a non-contact technique that utilizes images to convert the heat distribution data of a target object into a visual representation through measuring the infrared radiation emitted from the target object. This method allows researchers to surpass the limitations of human visual perception and observe the distribution of temperature on the object's surface. In the case of damage occurring inside a bridge component, the internal voids are frequently occupied by air or water. IRT is considered as an alternative imaging approach in order to detect and identify the type and location of subsurface defects [103].

Multiple research endeavors have successfully detected subsurface delamination through the analysis of temperature differences between the damaged and undamaged sections, which are unveiled through thermal imaging. For instance, Ali et al. [111] introduced a technique that utilizes thermal imaging and deep learning to identify subsurface damage in steel members of a truss bridge. In order to decrease computational expenses, the initial deep inception neural network (DINN) has been adapted for transfer learning. This approach offers bounding boxes to identify and locate subsurface damage, including corrosion and debonding between paint with coating and the steel surface.

Pozzer et al. [112] investigated the effectiveness of various deep neural network models in detecting primary concrete defects, such as delamination, cracks, spalling, and patches, in thermographic and conventional images captured from diverse distances and perspectives. These models were trained and evaluated using images obtained from a century-old buttress dam, and their accuracy was verified using images captured from two concrete bridges. The findings demonstrated that MobileNetV2 performs admirably in accurately detecting multiple forms of damage in thermal images.

Jin Lim et al. [113] suggested a hybrid technique for detecting corrosion in steel bridges that combines vision and IRT images. RGB images are captured using vision cameras, while active infrared amplitude images are obtained using infrared cameras. These two image types are merged to produce a hybrid image, which is then inputted into the Faster R-CNN algorithm for detecting corrosion damage on both the surface and subsurface of the steel bridge.

Tran et al. [114] presented an introduction to the current IRT techniques and case studies of passive IRT that have been employed on bridge construction projects. The investigation has shed light on the practical applications of IRT in identifying moisture penetration, insulation coverage, and subsurface delamination. Some guidelines have been recommended for employing passive thermography for on-site examination purposes. Lastly, a discourse is provided on common errors encountered during thermal surveying, aiming to assist engineers in mitigating avoidable errors.

In order to remotely evaluate the remaining thickness of the top coat in anticorrosion coatings on bridge surfaces, Sakata et al. [115] introduced a novel inspection approach based on near-infrared measurements. Their method enables quantitative assessment of the remaining thickness of the topcoat, regardless of the lighting conditions.

The majority of thermal imaging-based methods for identifying damage focus solely on determining the location and type of damage within the thermal image, offering only basic information about the damage. In contrast, damage segmentation methods go further by enhancing the results of damage detection, enabling the quantification of multiple parameters associated with the damage. Many of these techniques employ semantic segmentation networks in image processing. For instance, Ichi et al. [116] developed an

image-processing-based algorithm to detect subsurface delamination in concrete bridge decks using semantically segmented infrared images under varying ambient environmental conditions. By employing an adaptive image processing-based model, subsurface delamination was detected through the optimization of all user-defined parameters within the model. The optimization process contains the iterative selection of user-defined parameters and their impact on the performance metrics of the model.

Garrido et al. [117] presented the integration of a pre-processing algorithm and a Mask R-CNN Deep Learning model to detect severe defects from thermal images captured from bridges. The developed algorithm enhances the thermal contrast between areas with defects and those without defects in each individual image. Subsequently, the pre-processing algorithm outputs are employed in the training of Mask R-CNN to enable the automatic identification, segmentation, and classification of each region with defects.

Jang et al. [118] suggested a deep super-resolution segmentation network technique for evaluating cracks using a combination of vision and infrared images. This hybrid image-matching technique aims to improve the detectability of cracks while minimizing false alarms. To overcome the challenge of different image resolutions, a hybrid image-matching technique based on SrcNet was developed and tested on bridges in real-world conditions. By modifying the resolution of vision and infrared images, cracks at the 100 μm -level can be accurately assessed at the pixel level using hybrid image matching.

3.2.5. Hyperspectral Imaging

Hyperspectral imaging has been increasingly recognized as a valuable information resource for acquiring spatial and spectral characteristics, owing to its capacity to provide both RGB information and surface composition data based on wavelength-specific responses over wavelengths ranging from near-infrared to short-wave infrared [119].

Hyperspectral images have been employed in the examination of the coating thickness of the paint and degradation in steel bridge condition assessments. For instance, Alayoub et al. [120] utilized hyperspectral imaging and image classification approaches to establish a system for the monitoring of structural conditions, which is capable of rapidly evaluating the paint condition and the extent of degradation on large bridge structures. Ma et al. [121] applied hyperspectral images to evaluate the thickness of the paint coating on the steel based on the spectral pattern and detect various chemical binders and polymers in the coating layer. In another study, Mingyang et al. [122] proposed a technique that utilizes near-infrared to distinguish different coating conditions during the degradation process of steel structures.

Hyperspectral imaging has also been introduced as a noninvasive technique to identify steel corrosion. Lavadiya et al. [123] employed hyperspectral images to detect corroded surfaces and also identify the source of corrosion in steel bridges, which are difficult to extract through visual methods due to ambiguity that occurs in image classification/segmentation techniques. In a similar study, Ma et al. [119] proposed a method that uses hyperspectral reflectance indicators to monitor steel rebar corrosion that induces deterioration on bridge decks.

Furthermore, hyperspectral imaging can be applied to concrete bridge structures to analyze the material properties of concrete. As an example, Strauss et al. [124] proposed hyperspectral imaging analyses of concrete bridges to control the post-treatment in the structure and introduced this method as a reliable tool for lifecycle performance evaluation of concrete bridges.

3.2.6. Contact Sensors

In the majority of conventional techniques of bridge health monitoring, multiple sensors at various locations are installed on the structure, and the condition of a structure under normal operating conditions or extreme weather events is closely monitored by global indicators, such as vibration, and local indicators such as strains, or a combination of both.

In the realm of traditional contact-based sensing methods, the installed sensors typically measure dynamic structure data, such as acceleration, velocity, displacement, or inclination. Such sensors encompass a wide variety of instruments, including accelerometers, linear variable differential transformers, strain gauges, fiber optic sensors, piezoelectric sensors, impedance sensors, tele pendulums, and ultrasonic wave sensors. Therefore, several bridge damage detection methods based on structure sensors have been developed. Some of the studies used classification methods to handle the sensor data for bridge damage detection. For instance, Ghiasi et al. [125] proposed a deep learning approach based on vibration results to classify damages of different extents of losses in bridges. This approach utilizes CNNs that are trained on vibration data. Initially, field testing is conducted on a bridge using six accelerometers. Validation of a developed FEM of the bridge is conducted by the obtained modal parameters of the bridge. Following this, corrosion scenarios are simulated for the primary bridge members by introducing varying degrees of cross-sectional losses, as determined by the validated FEM. In the deep learning phase, a 1D CNN is developed, integrating innovative data augmentation techniques to precisely classify the acceleration responses corresponding to each simulated damage scenario. In another study, Parisi et al. [126] presented a technique for identifying structural defects in steel truss railway bridges by utilizing ML classification tools. This approach enables the analysis of unprocessed strain sensor signals, eliminating the need for any pre-processing or initial feature extraction. The data used in this study were produced through the simulation of various damage scenarios using finite element software. Subsequently, KNN and CNN classification tools were employed to process the generated data.

From another viewpoint, bridge damage detection using contact sensors can be solved using regression-based methods. For example, Malekjafarian et al. [127] suggested a two-step methodology for identifying bridge damage by employing an ANN trained with the measured vehicle responses obtained from multiple crossings over a healthy bridge. The vehicle response is anticipated based on its velocity during multiple crossings (monitoring dataset) across the bridge. Subsequently, a damage indicator is established using a Gaussian process to detect changes in the distribution of the forecast errors. Zheng et al. [128] presented a methodology for mitigating the impact of temperature fluctuations on the dynamic modal properties of bridges, which aims to enhance the detection of scour damage around bridge piles by employing vibration-based measurements. The proposed method utilizes a machine learning technique to establish the correlation between changes in modal properties and temperature variations, as measured by in situ sensors. Using Bayesian inference via Transitional Markov Chain Monte Carlo simulation, the modified vibration measurements, which effectively mitigate the impact of temperature variations, are employed to probabilistically infer the presence of bridge scours. Deng et al. [129] introduced a new approach to assessing the fatigue damage of the hanger without the need for direct stress sensing equipment. Their methodology involves utilizing data acquired from weigh-in-motion (WIM) sensors to gather traffic loading information, which is then combined with finite element analysis to estimate the daily fatigue damage of the hangers. Additionally, they employ Support Vector Machines (SVM) to establish regression models that correlate the daily fatigue damage with the collected traffic loading parameters.

From another point of view, structural sensor data can be transformed into an anomaly detection problem, which is frequently accomplished through the utilization of unsupervised learning or clustering techniques. The objective of the anomaly detection methods is to differentiate between regular and anomalous data that potentially result from bridge damages. As an example, Sarmadi et al. [69] introduced an anomaly detection approach in structural health monitoring that utilizes the adaptive Mahalanobis-squared distance and the one-class kNN rule (AMSD-KNN). To determine a precise threshold limit, the method employs the Generalized Extreme Value (GEV) distribution modeling using the block maxima technique. In order to select the most suitable block number, a goodness-of-fit measure based on the Kolmogorov–Smirnov hypothesis test is applied. Soleimani et al. [130] presented an unsupervised real-time SHM technique that employs a combina-

tion of low-and high-dimensional characteristics to train multiple ensembles of generative adversarial networks (GANs) and one-class joint Gaussian distribution models (1-CG). Furthermore, a detection system is established using the detection scores of GANs and 1-CG models for limit-state functions. The resistance of these limit-state functions is adjusted according to user-defined detection parameters by utilizing GAN-generated data objects through reliability analysis. The proposed method was applied to the Z24 Bridge dataset with vibration and environmental monitoring data. Entezami et al. [131] proposed a non-parametric approach for anomaly detection within the framework of unsupervised learning. This method uses the principles of empirical machine learning to establish a novel damage index, which is constructed through the utilization of empirical measures and the concept of minimum distance value. Initially, an empirical local density is computed for each feature, which is then multiplied by the minimum distance of said feature. This resultant product serves as the foundation for the newly derived damage index, which is beneficial for decision-making processes.

3.3. Digital Twin Data Connection Strategies

The communication tool within the DT implementation establishes a bidirectional connection that binds the physical assets to its virtual counterpart. In this section, the data connection role in DT will be highlighted through a review of the recent research works in this field.

The data connection in DT, as shown in Figure 5, is comprised of two functional blocks: the data transmission and data storage. The data transmission function supports the massive data exchange obtained from inspection and real-time monitoring on the physical bridge to the DT framework. It emphasizes the significance of this data flow mechanism in ensuring seamless integration and effective utilization of data within the DT environment. The data transmission must be able to transmit data with an acceptable latency and cost [22]. In addition, the data storage enables the massive data recording obtained from various sources during the bridge's lifecycle in order to facilitate seamless cooperation among different virtual models.

Regarding data transmission, the Internet of Things (IoT) serves as the primary facilitative technology that can seamlessly connect the digital replica with the physical asset and enables data mining and analysis due to its capability to gather data from diverse data sources with various transmission technologies and communication protocols [23].

Different data transmission technologies are supported by IoT devices. One of the most widely used data transmission methods is wired connections, such as Ethernet and Fieldbus, which is preferred by many DTs due to their lower costs and abundant bandwidth for fast and robust data transmission [41].

IoT-based Wireless Sensor Networks (WSNs) provide more flexibility in data transmission and facilitate the monitoring of bridges that are not accessible via cables. WSNs also provide continuous raw data from the bridge and play a crucial role in the DT generation [132].

WSNs consist of small sensors that transmit data through wireless connections. One of the key advantages of WSNs is their high energy efficiency and reliability, which make them well suited for real-time data collection scenarios [133]. In bridge monitoring, WSNs offer certain benefits over traditional wired sensors in terms of reliability and communication convenience. Using WSNs, it becomes possible to collect data from multiple cross-sections of the bridge, even in locations that are difficult to access [134].

Regarding the communication distance in WSNs, short-range wireless data transmission technologies, such as Wi-Fi, Bluetooth, Zigbee, or radio frequency identification (RFID), are appropriate for on-site data collection due to the limited communication distance using low-power radio frequencies. Short-range communication protocols provide reliable and secure data connections and enable devices to identify each other without the necessity of a central server or intermediary [135].

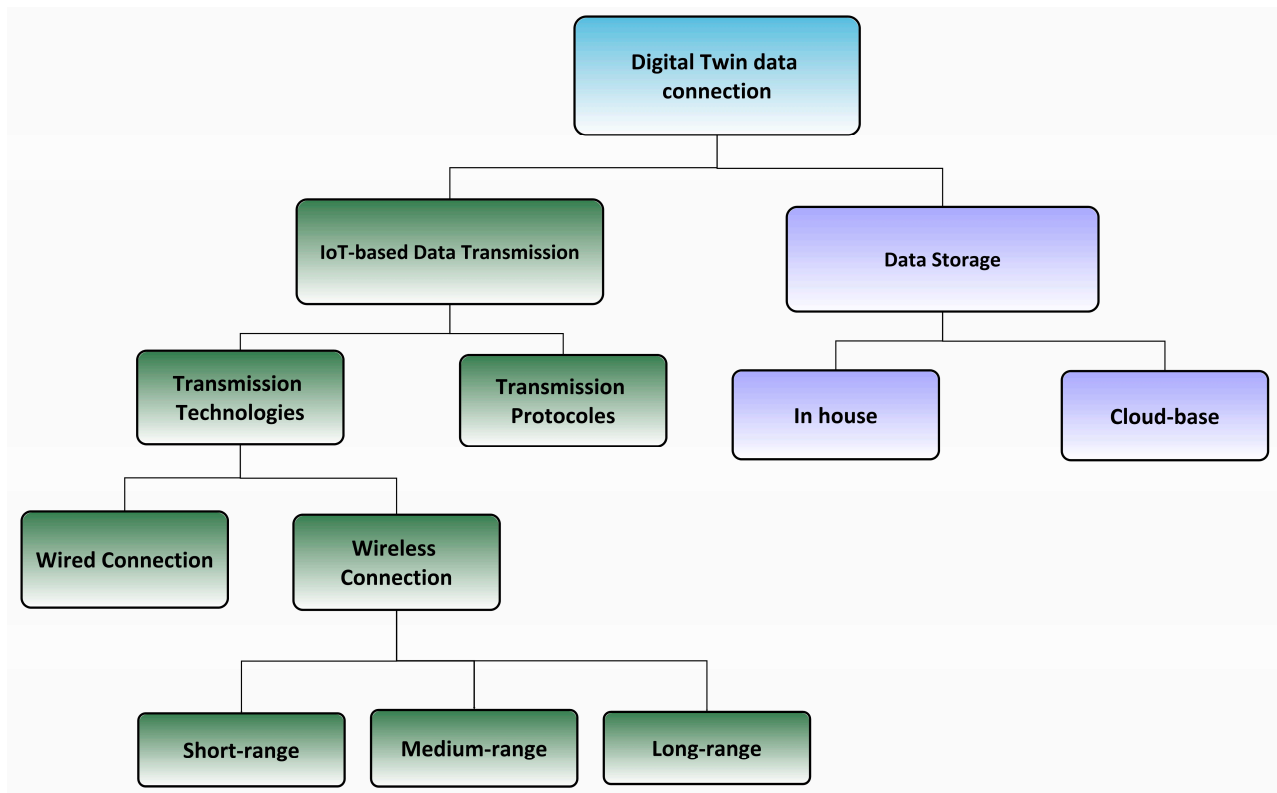


Figure 5. Overview of data connection methods in bridge digital twin.

Medium-range data transmission in bridge DT applications can be achieved using commercial cellular networks and low-power wide-area network (LPWAN) technologies. LPWAN technologies, such as NB-IoT and LTE-M, are appropriate for IoT applications since they offer networks with a considerable range and low power consumption [136]. Moreover, commercial cellular networks, such as 3G, 4G, and 5G, furnish higher rates of data transmission and support for seamless and high-speed connections [137].

Long-range technologies based on non-cellular LPWAN, such as LoRa, Sigfox, and Ingenu, are also considered as low-cost technologies that are suitable for monitoring of bridges in remote areas with limited resources. LoRa/LoRaWAN and Sigfox technologies have been developed for long-distance communication at a low energy and have obtained superiority over other LPWAN technologies in terms of the lifespan of devices, capacity of networks, adaptability of data rates, and cost-effectiveness [138].

The direct utilization of data collected from physical entities is not possible due to different data formats and, similarly, the data generated from virtual models cannot be directly used for a physical entity. Therefore, a proper data exchange protocol is required to enable the exchange of data between different IoT devices in bridge DT [139]. Message Queuing Telemetry Transport (MQTT), oneM2M, and Automation Markup Language (AML) are the most common and basic network communication protocols that are used in IoT-based devices [139].

Regarding the data storage concern, a common data environment, as well as a big data storage strategy, is required for the bridge DT in order to store an extensive amount of dissimilar data obtained from different data sources. The common data storage environment stores important bridge data, such as monitoring data, inspection reports, bridge design and construction documents, historical records, rules, and standards, as well as environmental data, such as traffic data, weather conditions, humidity, etc. [27]. BrIM with files in the Industry Foundation Classes (IFC) format is widely used to establish a common data storage environment due to its standard and shareable data layout [140]. Moreover, in cases where the information model in IFC format cannot capture the full complexity of real-world relationships, incorporating data into the FEM, geometric model,

and data model provides a more comprehensive representation that aligns with physical and operational aspects.

The concept of data storage in bridge DTs is closely intertwined with database technologies. However, as a result of the escalating volume and heterogeneity of multisource DT data, the traditional in-house database technologies are no longer practical. Consequently, big data storage technologies have gained significant attention, including distributed file storage (DFS), NoSQL database, NewSQL database, and cloud storage [73]. DFS allows multiple hosts to access shared files and directories simultaneously via the network. NoSQL is distinguished by its ability to scale horizontally in order to handle vast amounts of data. NewSQL refers to new databases that allow massive data storage and support ACID and SQL for traditional databases.

4. Classification and Investigation of Current Bridge Digital Twins

According to Boje et al. [141], the conceptual composition and overall structure of the DT varies according to the context of its field of application. In bridge engineering, due to the growing demands in the bridge engineering society for automated and intelligent administration systems, various fundamental components of bridge DT have been widely applied in diverse applications in bridge management and maintenance systems. Therefore, several recent studies have employed different viewpoints concerning the conceptual structure of DTs and adopted different perspectives for generating bridge DT. Honghong et al. [12] classified the current bridge DTs into two groups, pre-bridge DT and ideal-bridge DT, according to two aspects, including key performance parameters and functional requirements in bridge engineering. However, a comprehensive classification of bridge DTs is required to assist in determining the main functionalities of bridge DT. In this section, several factors, including applied models, data collection, functionality, objectives, and integrated technologies, have been considered to classify the bridge DT, as shown in Table 3. Two factors, including virtual model type and data collection methods, have been discussed in Sections 3.1 and 3.2, respectively. Other criteria are discussed in the following sections.

4.1. Main Objectives of Digital Twins in Bridge Management

The DT objectives refer to the all-encompassing goals or purposes that the implementation of a DT seeks to achieve. These goals include all the aims and potential outcomes that are expected from the deployment of DT technology within the bridge engineering field. These objectives can be summarized as below.

4.1.1. Monitoring

The majority of studies, as highlighted in Table 3, consider monitoring as a key application for the bridge DT by deployment of various sensor devices. The sensor network, which is positioned in the bridge structure, assumes a pivotal role in the acquisition of real-time data for day-to-day operational management. This procedure necessitates a meticulous selection and filtration of data in order to ensure that only pertinent information is taken into consideration. The aim is to present these data in a format that can be interpreted by machines. Consequently, AI systems or human operators utilize this information to make decisions concerning the virtual counterpart of the bridge [141].

4.1.2. Prediction

Prediction is another requirement for the bridge DT framework, aiming to foresee the future behavior and health status of the bridge infrastructure. Predictive modeling for structural life prediction underscores the immediate applicability of predictions to trigger actions on the physical side in response to anticipated events [142]. This predictive capability allows for proactive measures, enhancing the overall resilience and reliability of the bridge.

4.1.3. Simulation

Simulation is a core feature of the bridge DT, aiming to replicate the real-world scenario in a virtual environment with the highest level of fidelity. The simulation level and its precision are different based on the application and use case, necessitating adaptability in the DT platform [38]. In most case studies, FEM model updating is applied to analyze the structural integrity for bridge lifecycle monitoring. While this FEM simulation is feasible on smaller scales, challenges arise when dealing with larger structures [141].

4.1.4. Lifecycle Management

The importance of encompassing the entire lifecycle of the physical asset in bridge DT has been widely highlighted in different research studies to reduce the long-term remediation costs [37]. The approach to planning for the lifecycle naturally differs based on the application domain. For bridge infrastructures, lifecycle management is regarded as an ongoing process that optimizes running costs, structural integrity, and safety [143].

4.1.5. Decision Support System (DSS)

Decision support has yet to be integrated into bridge DTs. It encompasses the fusion and analysis of data from the monitoring, prediction, and simulation phases, forming the basis for well-informed decision-making. The collaborative interplay between AI systems and human operators is central to the decision-making process within the DT. Decision support operates on both real-time and long-term scales, ensuring proper responses to immediate issues while facilitating strategic planning for the entire lifecycle of the bridge [8]. The adaptability of decision support systems to dynamic conditions, coupled with its focus on risk mitigation and cost optimization, enhances the resilience and sustainability of the bridge infrastructure [144].

Bridge DT decision support primarily enables the planning strategies for the inspection, maintenance, repair, and rehabilitation tasks associated with bridges. Therefore, DT decision support can offer valuable guidance for decision-making in bridge maintenance at the bridge-component level. These decisions involve the selection of specific components for repair and rehabilitation, with the aim of minimizing lifecycle costs, avoiding delays in repair that could lead to component damage and failure, and maximizing the overall structural condition [41].

Moreover, DT decision support can ease the network-level decision-making for bridge authorities. At the road-network level, decision-making prioritizes maintenance and repair tasks for the most critical bridges in the road network. The decision-making approaches employed are comparable to those utilized for bridge-level decisions. However, the objectives can encompass the minimization of overall costs and maintenance delays associated with all bridges, as well as the maximization of network performance, which may involve the reduction of total travel time and distance and the enhancement of network connectivity [41].

4.2. Digital Twin Functionality

As bridge DT serves as a dynamic, virtual counterpart to its physical counterpart, studies in this field have considered a spectrum of functionalities for bridge DT that collectively contribute to a holistic approach to meet DT objectives (Section 4.1) throughout the bridge lifecycle. These functionalities refer to the specific computational methods and tools embedded in the DT system to operate and replicate, simulate, and interact with the physical part. These functionalities can be summarized as follows:

- **FEM Model Updating:** This functionality entails the continuous refinement of the DT's representation through integration of sensor data streams in FEM models. This refinement is essential for detecting damage because it allows for quantifying parameter changes necessary to update a baseline FEM model to match data from a damaged state. This alignment between the model and data is fundamental for accurate damage detection and assessment in structural health monitoring systems [24–28].

- **As-is 3D Model Updating:** This functionality focuses on the continuous synchronization of the 3D representation of the DT with the present physical condition of the bridge. As the bridge undergoes changes or natural deterioration happens with its operational life, the as-is model updating replicates these modifications. The synchronization supports better decision-making for maintenance and repairs, as the DT provides an up-to-date and detailed understanding of the bridge's current state [29–32].
- **Data-Driven Model Updating:** This functionality focuses on providing the DT's representation by the use of AI-based approaches. By leveraging real-time insights and analyzing vast amounts of data using machine learning algorithms, the DT can forecast optimal intervention points and preclude potential issues before they manifest. This proactive approach, based on identifying patterns, anomalies, and trends, enhances maintenance strategies and improves operational efficiency by addressing potential issues before they become critical [39,145].
- **Decision-Making:** This functionality actively participates in informed decision support processes related to maintenance, repairs, and potential upgrades of the bridge and aids in recommending optimal strategies for maintenance and enhancements based on current and predictive data. This functionality leads to more cost-effective and sustainable asset management practices, ensuring that resources are utilized efficiently while maintaining the bridge's functionality and safety [22,146].
- **Virtual Models' Integration:** This functionality orchestrates the seamless amalgamation of the different virtual models within the DT's environment. By integrating virtual models, the DT provides a comprehensive and interconnected understanding of the bridge's current and future condition. This integrated approach facilitates better collaboration among stakeholders, improves risk assessment capabilities, and supports long-term planning for bridge management [12].

4.3. Integrated Technologies

The incorporation of advanced technologies has become instrumental in the revolution of bridge digital twins to facilitate its implementation. Through the seamless integration of cutting-edge technologies, engineers have the ability to construct comprehensive digital replicas of bridges, which not only could provide real-time observations, predictive analytics, and improved decision-making capabilities, but also empower bridge managers with a deeper understanding of their behavior, performance, and resilience in the face of dynamic environmental conditions and operational demands. The technologies most widely used in bridge DTs can be summarized as below.

4.3.1. Internet of Things

The IoT-based sensing technology has been included in bridge DT, as it plays a crucial role in constant bridge monitoring and data collection using internet-connected sensors and devices. By combining real-time data obtained from IoT devices with DT technology, a high-level depiction of bridges can be created, allowing the gap between the physical and virtual worlds to be filled [147]. The integration of IoT devices in bridge DT also facilitates massive inspection data transformation from restricted communication areas and supports timely DT services, as investigated by Gao et al. [27] through development of an AIoT-informed DT communication framework to solve the time delay of bridge DT services according to communication and computation complexity.

4.3.2. Artificial Intelligence

AI is considered another crucial part of most bridge DTs, contributing significantly to their predictive and optimization capabilities [148]. The term AI covers a wide range of approaches, which include machine learning, data mining, logics-based AI, and knowledge-based AI. This variety highlights the multifaceted character of AI's impact on DTs [16]. AI techniques facilitate higher-level computations within bridge DT and enable to dynamically process and analyze vast amounts of data and predict outcomes, automatize processes, and

make smarter and more informed decisions. This dynamicity is crucial for the real-time adaptability of DTs, allowing them to go beyond static representations and achieve higher levels of sophistication and intelligence [148].

4.3.3. BrIM

BrIM has been widely used in different phases of design, construction, operation, and maintenance phases, and many BMSs have been developed in the literature based on BrIMs. Moreover, BrIM is considered as the starting point of bridge DTs, serving as a semantically reference model enriched with costs, time, sustainability, and 3D data. Therefore, several research studies tried to combine the BrIMs, as an important source of data, with DT using different technologies, such as UAVs [33,35], TLS [8], holographic projection [36], and extended reality [149].

4.3.4. Data Exchange and Integration

The integration of different digital technologies in bridge DT necessitates a thorough consideration of a seamless data exchange strategy to provide an opportunity to integrate diverse data sources, such as bridge design and dynamic sensor data, to create a comprehensive representation of the bridge. Static and real-time data can be connected through linked data technologies and system integration approaches, allowing for data-driven applications and uncovering hidden inefficiencies [150]. Additionally, linked data facilitate the merging of data from different systems, making it possible to create a DT by utilizing existing systems. As shown by Hakimi et al. [151], the Industry Foundation Classes (IFC) is considered as a well-established and standardized data model that ensures interoperability across diverse software platforms.

4.3.5. Expert Knowledge

Expert knowledge plays an important role in ensuring the accuracy, reliability, and effectiveness of the bridge DT. It aids in generating accurate and reliable digital replicas of physical bridges by providing insights into the behavior and characteristics of real-world structures. Expert knowledge is used to design and optimize digital simulation models that can accurately predict and simulate the behavior of the physical bridge in the virtual space [75]. It also aids in evaluating and managing the complexity of the system design stage by integrating DT with model-based systems engineering. Furthermore, expert knowledge is utilized in combining bridge visual inspection results and finite element modeling to create DTs of bridges, which can be used for bridge design, management, and operation [152].

Table 3. Assessment of current research studies on the bridge digital twins.

No.	Reference	Virtual Model Type	Data Collection \ Sensing Method	DT Main Functionality	DT Data Representation Method	Main Objectives of DT					What Is Considered as a Part of DT?			
						Monitoring	Prediction	Simulation	Lifecycle Management	Decision System	IoT	AI	BrIM	Data Integration
1	[31]	FEM-based model	Sensor data	FEM Model updating	FEM model	✓	✓	✓	-	-	-	-	-	✓
2	[153]	FEM-based model	Sensor data, NDT survey	FEM Model updating	FEM model	✓	-	✓	-	-	-	-	-	✓

Table 3. Cont.

No.	Reference	Virtual Model Type	Data Collection \ Sensing Method	DT Main Functionality	DT Data Representation Method	Main Objectives of DT					What Is Considered as a Part of DT?				
						Monitoring	Prediction	Simulation	Lifecycle Management	Decision System	IoT	AI	BrIM	Data Integration	Expert Knowledge
3	[30]	FEM-based model	Sensor data	FEM Model updating	FEM model	✓	-	✓	-	-	-	-	-	-	✓
4	[145]	FEM-based model	Crack information	FEM Model updating	FEM model	✓	✓	✓	-	-	-	-	-	-	✓
5	[28]	FEM-based model	Sensor data	FEM Model updating	-	✓	-	✓	-	-	-	-	-	-	✓
6	[29]	FEM-based model	Sensor data	FEM Model updating	FEM model	✓	✓	✓	-	-	-	-	-	-	✓
7	[154]	FEM-based model	Sensor data, RGB images	FEM Model updating	3D model	✓	-	✓	-	-	-	✓	-	-	-
8	[63]	Geometric 3D-based model	Point cloud	As-is 3D Model updating	BrIM	✓	-	-	✓	-	-	-	✓	-	✓
9	[146]	Geometric 3D-based model	RGB Image data	As-is 3D Model updating	BrIM	✓	-	-	-	-	-	-	✓	-	✓
10	[155]	Geometric 3D-based model	Point cloud	As-is 3D Model updating	3D Model	✓	-	-	-	-	-	-	-	-	✓
11	[156]	Geometric 3D-based model	RGB image data	Decision-making	3D Model	✓	-	-	-	✓	-	-	-	-	✓
12	[8]	Geometric 3D-based model	Point cloud	As-is 3D Model updating/Decision-making	BrIM	✓	-	-	-	✓	-	-	✓	-	✓
13	[157]	Data-driven model	Sensor data	Data-driven Model updating	FEM model	✓	✓	✓	-	-	-	✓	-	✓	✓
14	[43]	Data-driven Model	Sensor data	Data-driven Model updating	-	✓	✓	✓	-	-	-	✓	-	✓	-
15	[75]	Data-driven model	Sensor data	Data-driven Model updating	-	✓	✓	✓	-	-	-	-	-	✓	✓
16	[158]	Data-driven Model	Sensor data	Data-driven Model updating/Decision-making	BrIM	✓	-	-	-	✓	-	✓	✓	-	✓
17	[27]	Data-driven Model	Sensor data, RGB images	Data-driven Model updating/Decision-making	3D model	✓	✓	✓	-	✓	✓	✓	-	✓	✓
18	[159]	Data-driven Model/FEM-based model	Sensor data	FEM Model updating	BrIM	✓	✓	✓	-	-	-	-	✓	-	✓

Table 3. Cont.

No.	Reference	Virtual Model Type	Data Collection \ Sensing Method	DT Main Functionality	DT Data Representation Method	Main Objectives of DT					What Is Considered as a Part of DT?			
						Monitoring	Prediction	Simulation	Lifecycle Management	Decision System	IoT	AI	BrIM	Data Integration Expert Knowledge
19	[22]	Data-driven Model/FEM-based model	Sensor data	FEM Model updating	3D model	✓	✓	✓	-	-	-	✓	-	✓
20	[12]	All Models	Sensor data, RGB image, Point cloud	Virtual Model Integration	AR	✓	✓	✓	✓	-	✓	✓	✓	✓

As the results of Table 3 show, most of the bridge DTs in the literature focus on bridge monitoring using FEM model updating to simulate the current condition and predict the future status of the bridge [160]. These methods combine FEM, data-driven virtual models, and bridge health monitoring results to create a DT.

However, as the results of Table 3 and keyword co-occurrences analysis (Section 2.2) show, integration of Decision Support Systems (DSS) in bridge management systems has been rarely considered in bridge DTs. They can provide valuable tools for enhancing maintenance efficiency, improving decision-making efficiency, and reducing decision-making time and costs. Therefore, they can be positioned at the heart of DT to synthesize the data gleaned from monitoring, predictions, simulations, and lifecycle management into actionable insights.

Most of the current DTs developed for managing bridge infrastructure assets do not incorporate information from all stages of the asset's lifespan. The successful management of assets throughout their lifespan necessitates the consistent digital flow of data across different stages of the bridge lifecycle, thereby enabling the prediction of bridge performance and the formulation of intelligent maintenance decisions.

5. The Proposed Reference Framework for Bridge Digital Twin

Based on the review presented, this research study presents an evolved bridge DT framework designed to manage data flow across different stages of the bridge lifecycle. This framework utilizes the essential components developed in current DT methods [12,16,22,151] (Section 4) and effectively integrates data obtained from different data collection methods and virtual models. The general concept of the proposed framework, shown in Figure 6, includes an additional intelligent Decision Support System component.

The proposed framework operates through different data collection methods, and the data connection step provides a sole repository for information gathered from diverse methods. The acquired data undergo processing and analysis using different virtual models, enabling multi-level data fusion, which integrates information from various sources for optimization decision-making processes. The intelligent decision support incorporated in the proposed framework plays a pivotal role in the optimization of decision-making processes. Harnessing the power of advanced algorithms and machine learning models, the Decision Support System excels at extracting meaningful insights from the amalgamated data repository. Its ability to discern patterns, trends, and anomalies equips it to provide invaluable recommendations for informed decision-making at multiple levels. This comprehensive approach empowers the Decision Support System to predict potential

issues, recommend preventive measures, and optimize maintenance schedules. While the proposed framework primarily enhances decision support during the operation and maintenance phase of bridge management, it also positively impacts decision-making processes during the planning, design, and construction phases of the bridge.

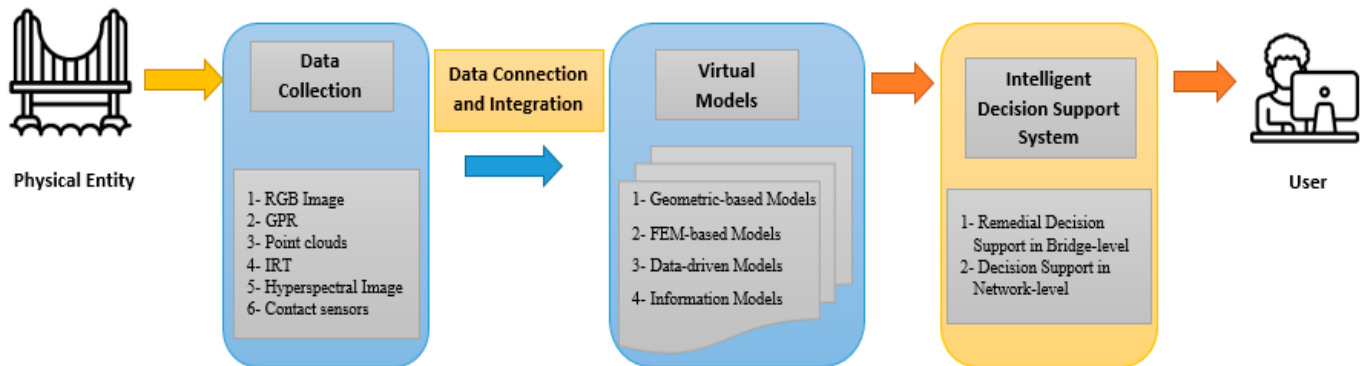


Figure 6. The overall concept of the proposed bridge DT framework.

The innovations of the proposed framework include a combined data collection method through a reliable data storage layer for diverse data sources, enabling advanced multi-source data fusion. Moreover, this framework integrates different virtual models through a model fusion process, enhancing the ability to precisely assess the current condition of bridge. In addition, an intelligent decision support layer is integrated in this framework, which empowers the system to optimize maintenance schedules effectively. This framework integrates a variety of technologies, including DT modeling technologies, big data, machine learning, IoT, augmented reality (AR), cloud computing, communication technologies, and more. Each technological component within the framework can be implemented using a range of tools, as comprehensively reviewed by Rathore et al. [16].

The proposed framework is structured into several layers, as shown in Figure 7, including a monitoring and data collection layer, a data transfer layer, a data processing and storage layer, a digital twining layer for data analysis, an intelligent DSS layer, and a visualization/control layer that includes visualization of DT and provides an automatic control platform. At distinct stages of the process, additional checks and validations are carried out. Machine learning methods are utilized to detect anomalies in IoT data, while expert assessments are enforced to maintain human oversight and integrate professional judgment into essential decision-making phases. AR is also employed to aid in decision visualization, interactive engagement, and remote control, enhancing the overall functionality of the system.

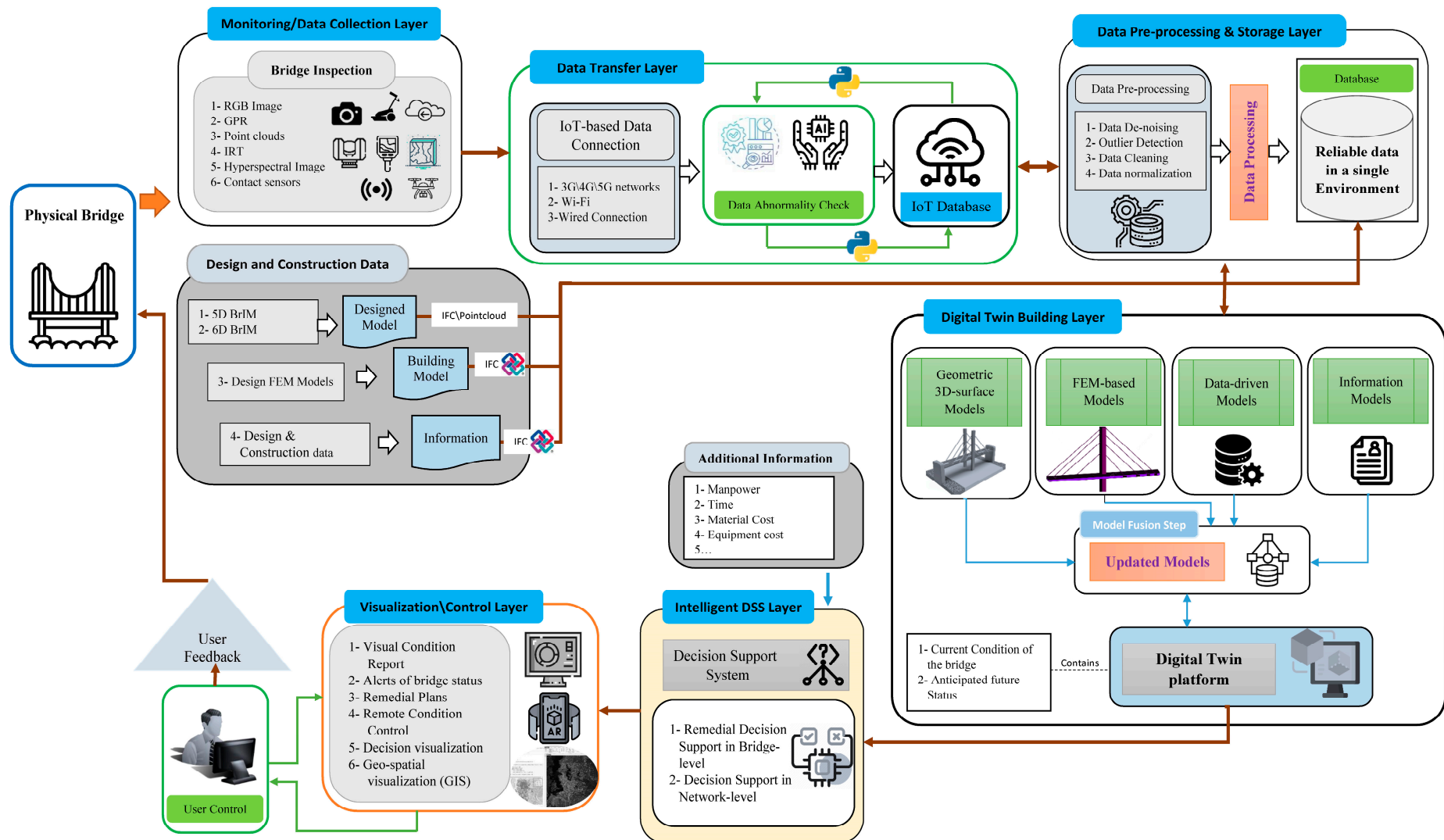


Figure 7. The evolved bridge digital twin framework.

5.1. Monitoring and Data Collection Layer

The data collection process is the initial part of DTs and begins with bridge inspection using various sources/tools, as explained in Section 3.2. Different types of contact or non-contact sensors with various technologies may be used to extract specific information about the bridge's structural health status. These technologies are crucial for both operational condition assessment and as-built/as-is model creation. A diverse range of tools facilitate the collection of data for DT, including UAVs, TLS, LiDAR, and photogrammetry tools.

Data can be obtained either offline or in real time. Real-time monitoring allows for the capture of dynamic changes and variations, providing a more accurate representation of the bridge's behavior under different conditions. However, in offline data collection, data are gathered and recorded locally on devices, such as laser scanners, cameras, or GPR. Although this type of data collection might avoid real-time analysis and decision-making, it provides a remote, flexible data collection and inspection, which is geometrically accurate and of superior quality.

5.2. Data Transfer Layer

Regarding data transmission, IoT-based data connection is primarily used for bridge data collection, leveraging IoT devices and technologies to collect, transmit, and integrate real-time data from various sources within an optimized process. Furthermore, IoT streamlines the automation of data acquisition and integration by utilizing sensors to gather real-time data from physical asset and share them with other layers through different kinds of wired or wireless communication and data transmission methods. Such integration promotes a bridge management approach that is more responsive and adaptive.

Data collected from monitoring sensors are transmitted to a cloud-based IoT database. To maintain the quality and precision of IoT data, learning-based verifications are conducted on the database to validate input data and detect abnormalities. These algorithms are trained to collaboratively validate data from various collection instruments and detect anomalies in the recorded data.

5.3. Data Pre-Processing and Storage Layer

The large amount of real-time data will be pre-processed using different techniques, such as noise removal, outlier detection, data cleaning, and normalization, which is crucial to ensure the integrity and accuracy of the data. Moving on to the processing of offline data, various sophisticated techniques are employed to extract pertinent information about the bridge's current condition from these diverse data sources. Image processing algorithms are applied to RGB and IRT images to identify structural deformations or anomalies, providing visual insights into potential issues. Hyperspectral analysis aids in identifying material composition and detecting subtle changes that may indicate structural weaknesses. Point clouds contribute to creating a detailed 3D representation of the bridge, enabling a comprehensive evaluation of its geometry and any potential deformations. GPR data are processed to identify subsurface anomalies or structural changes that may not be visible through other methods. The integration of the results obtained from offline data processing with the real-time data from contact sensors ensures a holistic understanding of the bridge's condition, combining the strengths of various data types for a thorough assessment.

The integrated data are consolidated within a central database, which serves as the central data hub, ensuring a cohesive storage structure for reliable inspection of data in a single environment. Consequently, all pertinent information regarding assets can be readily accessed from this data hub within the DT framework. Combined with design and construction data, and historical inspection and asset management records, this single database hosts a comprehensive spectrum of information that improves the collaboration and decision-making processes in the next step.

5.4. Digital Twinning and Model Fusion Layer

Virtual models play a crucial role in various aspects of bridge DT, and different types of models are used to simulate, represent, and analyze different aspects of a bridge. As explained in Section 3.1, different virtual models can be used, including 3D surface models, simulation FEM models, data-driven models, and information models.

The 3D surface models provide a detailed 3D representation of the bridge's geometry. These models capture the surface condition and spatial characteristics of the bridge, including its shape, dimensions, and surface features. The process of generating 3D surface models involves cutting-edge techniques, such as 3D laser scanning or UAV photogrammetry, to create a highly detailed and accurate point cloud. This point cloud is then transformed into a 3D surface model, providing an accurate digital replica of the bridge. The resulting 3D surface models become invaluable for structural assessments and condition monitoring.

Simulation-based FEM modeling entails generating virtual representations using the actual geometric dimensions and material properties of the bridge through finite element software. These representations can simulate various scenarios, including traffic loads, environmental conditions, or seismic events. By running simulations, the bridge responses to various conditions and potential issues could be analyzed and structural integrity could be assessed through identifying potential weak points. The bridge's dynamic behaviors can be studied using stress distribution, deformation patterns, and resonance effects, allowing for the identification of critical areas that might require attention or reinforcement.

Data-driven models are computed through AI-based mathematical models that approximate the behavior of the bridge based on exploration of the relationship between data. These models can be used to predict the bridge's response under different conditions without the need for extended and time-consuming simulations or physical testing. Edge computing has the capability to offer expedited predictive maintenance determinations by carrying out predictive analytics in close proximity to the data origin, thus enhancing the analytics process speed and introducing scalability to the overall system [161]. In bridge DT, edge computing enables the analysis of real-time data, which facilitates immediate decision-making capabilities. In addition, Topological Data Analysis (TDA) [162] is another powerful approach that can be harnessed as a data-driven method. TDA focuses on understanding the shape and structure of data, especially in high-dimensional spaces. By leveraging techniques from algebraic topology, TDA can reveal patterns, clusters, and relationships within complex datasets that may not be easily apparent through traditional methods.

Information models go beyond geometry and include a wide range of information related to the bridge's lifecycle. This can include design data, construction details, material specifications, maintenance schedules, and more. Information models facilitate collaboration among various stakeholders and support decision-making throughout the bridge's lifecycle.

5.5. Intelligent Decision Support Layer

The ideal DT should offer practical functionalities to bridge managers and decision-makers to continuously control, monitor, and optimize the physical bridge. In a case where maintenance interventions are necessary, a decision support layer in DT should enable an appropriate scheduling and prioritization of the maintenance actions through data-driven, updated, and accurate procedures. In addition, notwithstanding the extensive utilization of DT to assist decision-makers by furnishing insights into the structural reliability of an asset when it is operating under normal conditions, the significance of the DT decision support layer would be more pronounced subsequent to exceptional occurrences, such as earthquakes or floods. With the information provided by the decision support layer and the aid of predetermined thresholds and emergency plans, decision-makers can effectively handle emergency situations, facilitating prompt and efficient responses. When all of these factors are taken into consideration, it becomes evident how DSS can be seamlessly integrated into a comprehensive DT for bridge management.

Due to the inherent contradictory nature of the goals associated with bridge maintenance, the task of making optimized decisions is usually fraught with difficulty. Therefore, decision-making becomes a complex problem that requires the consideration of multiple objectives in order to achieve the best solution. Various optimization techniques can be employed to identify the most optimal solution, such as evolutionary algorithms in different forms of genetic algorithms [163], and multi-objective optimization with deep learning [164].

Additional investigations that can be considered by the DSS layer in order to optimize the decision-making are the expenses associated with maintenance, repair, and rehabilitation. These investigations include the costs of routine repairs and inspections, the costs incurred when replacing failed components, as well as the social costs, such as the increased travel time and distance experienced by users when the bridge is out of operation, and the costs associated with fatalities resulting from bridge failures and environmental costs [165].

5.6. Visualization and Control Layer

The outcomes of the analysis conducted in the decision support layer will be displayed using a user-friendly visualization layer. This layer converts intricate analysis findings into easily understandable visuals, guaranteeing that stakeholders can effortlessly comprehend and interpret the information. Decisions made by the DT system will be subject to human oversight to maintain the integral involvement of humans in the decision-making process.

The visualization and control layer is equipped with a remote condition control section, which is a key feature that enables continuous monitoring and management of the bridge's condition. This capability ensures that stakeholders can stay informed and take necessary actions, even without physical presence at the bridge site. In addition, it is capable of generating real-time alerts of bridge status, promptly notifying stakeholders of any critical changes or emerging issues. These alerts serve as proactive indicators, allowing for swift responses and preventive measures to maintain the bridge's integrity. The decision visualization component in this layer enhances the system's efficacy by providing an intuitive representation of the decision-making process. This visual insight into the rationale behind decisions empowers stakeholders to understand, trust, and contribute to the overall management strategy effectively, and if remedial action is required, the development of remedial plans is shown.

In order to enhance the visualization and integration of the decision within the real-world context, augmented reality (AR) is utilized for the purpose of decision visualization, interactive engagement, and remote control. AR enables decision visualization by overlaying virtual information onto the real-world environment. This functionality empowers users to perceive pertinent information, including simulations, models, and data analytics, integrated seamlessly into their surroundings via AR-enabled devices, such as smartphones, tablets, or AR glasses. This technology allows decision-makers to visualize complex scenarios, designs, or data in real-time and in context with the actual environment.

Moreover, the GIS map is utilized to locate the bridge geographically in order to incorporate spatial data into the visualized model. The data from the information model of DT are instantaneously transmitted to the GIS map, and any changes in the condition of the bridge will be promptly updated in the GIS-based attribute tables for visualization and spatial information querying. This real-time synchronization holds immense practical value, which helps stakeholders to rely on an up-to-date GIS representation that mirrors the current state of the bridge.

6. Challenges in Proposed Framework Implementation

The proposed framework of bridge DT revealed distinct evolutionary developments in bridge DT, showcasing advancements in key aspects, including: (1) advancement in offline and real-time data acquisition and fusion, (2) application of cutting-edge technologies for data management and processing, (3) enhanced virtual model integration and interaction,

(4) advanced intelligent decision support, and (5) immersive visualization and control capabilities integrating user feedback.

To realize the sophisticated advancements of the proposed bridge DT framework, it is crucial to overcome some implementation barriers. The challenges can be condensed into the following:

1. The maintenance of real-time updates for DT virtual models presents a significant challenge in the bridge DT technology. A robust data connection is required to efficiently transmit data from sensors installed on the physical bridge to the computer systems, to ensure that virtual models accurately reflect the reality, which is a challenging part of bridge DT. Furthermore, the complete automatic utilization of data acquired from UAVs and offline inspection tools for updating DT models remains unrealized, despite their impressive data-capturing capabilities. These tools rely on significant human resources and time for data preparation, processing, and computation of outcomes, resulting in significant constraints in real-time model updating.
2. The development of more practical AI-based techniques is crucial for fully leveraging the potential value of virtual model integration and maximizing the efficiency of the proposed bridge DT framework, despite the considerable work that has already been done in this area.
3. An efficient data integration and exchange standard is required for the proposed framework. Although the IFC format is regarded as a well-known technological advancement, further advancements are required to extend its capabilities to prevent potential data loss during transitions between different sections in bridge DTs.
4. As Decision Support Systems have not been widely considered within the existing bridge DTs, their implementation in the proposed framework does present a set of challenges, including the data quality and accuracy impact, responsive and prompt decision achievement with real-time data, and intuitive human-machine interactions for expert control.

7. Conclusions

This paper investigated the latest technological advancements and research trends on DTs within the context of bridge engineering through a scientometric analysis conducted on more than 480 published research documents. The literature review presented a comprehensive analysis and classification of the existing DT components and functional characteristics for bridge engineering.

Additionally, the paper thoroughly analyzed existing DT definitions and classification concepts, identifying key performance indicators for DT functionality. It hierarchically classified the objectives, functionality, and integrated technologies of DTs specific to bridge engineering. Through the extensive literature review study, this paper concluded that decision support strategies in management systems have rarely been considered in bridge DTs, and it can help to fully explore the inherent values associated with bridge DTs in bridge engineering.

Furthermore, the paper presented an evolved multilayer framework for DTs customized for full-lifecycle bridge management. The proposed framework provides a conceptual development in bridge DT and integrates different objectives, functionalities, and implementation procedures, designed to address the entire lifecycle of bridge management. The primary objective of this framework is to enhance accurate data collection, digital twinning and model fusion, reliability of remedial decisions, and visualization in bridge engineering. This framework is anticipated to make an active contribution to the future development of DTs through different layers that integrate advanced technologies.

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